



Pathways to scaling up in emerging economies: A configurational analysis of organizational capabilities in social enterprises

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ABSTRACT

Organizational capability is a key factor influencing the effectiveness of scaling social impact within social enterprises, yet the combined effects of specific capabilities have been largely overlooked by existing studies. Utilizing the SCALERS model, this research examines how different configurations of organizational capabilities contribute to high-performance scaling up in entrepreneurially vibrant regions of emerging economies. Based on a configurational analysis of 46 purposively selected Chinese social enterprises, this study identifies five effective pathways catering to social enterprises with varying replicable programs, connection capacities, and self-development capabilities. These pathways remain robust when government attention and public attention are considered situational contingencies. Additionally, this study reveals a high level of consistency across pathways in achieving overall social impact and its specific quantitative and qualitative dimensions. This study bridges the gap in examining the synergistic potential of organizational capabilities and provides a more fine-grained investigation of scaling up.

1. Introduction

Social enterprises are organizations that employ business-driven solutions to advance social objectives (Haugh, 2006). Aiming to catalyze large-scale, systemic, and sustainable social changes, social enterprises strive to increase the reach and intensity of their impact – a process commonly known as “scaling up” (Dacin et al., 2010; Islam, 2020; Lyon & Fernandez, 2012). The scale of social impact achieved through scaling is considered the foremost measure of a social enterprise’s effectiveness (Ebrahim & Rangan, 2014), making it a central concern for both practitioners and scholars in the field (Islam, 2020; Lyon & Fernandez, 2012). A key determinant of scaling success, anchored in the resource-based view, is organizational capability – a fundamental capacity to mobilize resources and undertake activities (Amit & Schoemaker, 1993; Helfat & Peteraf, 2003; Inan & Bititci, 2015). By developing multi-dimensional organizational capabilities, social enterprises are viewed as able to leverage their internal strengths and respond to external conditions, enhancing their performance in scaling up (Bacq & Eddleston, 2018; Bloom & Chatterji, 2009; Urban & Bukula, 2022).

The systematic conceptualization of organization capability in social enterprises was introduced by Bloom & Chatterji (2009) through the SCALERS model, which defines seven interrelated dimensions: staffing,

communicating, alliance-building, lobbying, earnings generation, replicating, and stimulating market forces. Since its inception, the SCALERS model has become the leading theoretical framework for examining how social enterprises scale, with widespread applications across contexts such as the USA, Italy, and China (Bloom & Smith, 2010; Cannatelli, 2017; Day & Jean-Denis, 2016; Yu et al., 2022). Built on contingency theory and the ecosystem approach, this model posits that the efficacy of each capability – and their combinations – varies by circumstance, suggesting that not all capabilities are equally essential for successful scaling and that multiple paths to success exist (Bloom & Chatterji, 2009).

Despite this emphasis on synergies and contingencies, existing research has primarily examined the impact of these capabilities separately, largely overlooking their interplay and configurational effects on scaling effectiveness. Given the resource constraints commonly faced by social enterprises (Bacq & Eddleston, 2018; Day & Jean-Denis, 2016), identifying which organizational capabilities are essential and which combinations are most likely to drive high-performance scaling is crucial. To address this gap, this study employs Qualitative Comparative Analysis (QCA) to investigate the causal relationships between configurations of SCALERS capabilities and scaling performance in social enterprises.

QCA is renowned for its ability to explore complex causal

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relationships with synergistic and asymmetric effects, but it requires a high degree of homogeneity in unchosen causal variables (Rihoux & Ragin, 2008). Therefore, this study focuses on entrepreneurially vibrant regions in emerging economies to ensure uniformity in the entrepreneurial environments observed. This choice is driven by two practical considerations: first, the intricate social challenges and institutional voids in these economies have spurred a rise in social enterprises, yet empirical research in this context remains scarce (Holt & Meldrum, 2019); second, social enterprises in developing countries tend to cluster in regions with more developed entrepreneurial resources (Iskandar et al., 2023; Jenner, 2016). Concentrating on this context not only enriches empirical investigations into social entrepreneurship in emerging markets but also broadens the applicability of findings to a wider range of social enterprises within these settings.

China is an exemplar of emerging economies, epitomizing the development of social entrepreneurship within such contexts (Sengupta et al., 2018). This study selected 46 social enterprises from various Chinese cities, identified as vibrant entrepreneurial environments through professional indices. Initially, the research assesses whether certain organizational capabilities serve as necessary conditions for (not-)high-performance scaling. Subsequently, it ascertains which capability configurations effectively drive high-performance scaling. Recognizing that scaling involves both qualitative (depth) and quantitative (breadth) impact (Desa & Koch, 2014), the study conducts comparative analyses across three dimensions: overall, depth, and breadth impact. Lastly, a supplementary analysis is incorporated to test the robustness of the identified configurations within complex institutional settings, examining how two crucial situational contingencies – government and public attention to issues engaged by social enterprises – affect the pathways.

This study contributes to the literature in three key ways. First, beyond responding to calls for more research on scaling social impact (Shepherd & Patzelt, 2022), it provides detailed insights into the qualitative and quantitative dimensions of scaling with targeted management implications. Second, it addresses a critical gap in understanding the synergy and contingency of organizational capabilities, providing empirical support for the theoretical claims of the SCALERS model. Finally, it highlights the potential of configurational analysis for exploring complex causal relationships in social entrepreneurship, presenting an operational example for broader application.

The next section reviews key concepts related to the research questions, including their definitions and relevance. This is followed by the methodology section, detailing the analytical approach, case selection, data collection, variable measurement, and calibration. The results section then presents findings from the necessity conditions analysis, sufficient solutions, and supplementary analysis. The discussion section interprets the theoretical contributions and practical implications of the primary findings. The article concludes with a reflection on research limitations.

2. Theoretical background and literature review

2.1. Theoretical background

This study is grounded in the theoretical framework of the SCALERS model, initially introduced by Bloom & Chatterji (2009) to analyze scaling social entrepreneurial impact through the lens of organizational capabilities, with further theoretical and empirical enhancements by Bloom & Smith (2010). This model's theoretical contributions are threefold: first, it conceptualizes organizational capability in social entrepreneurship organizations as seven distinct capabilities and demonstrates their relevance to scaling both theoretically and empirically. Second, it recognizes the key situational contingencies that influence these capabilities, underscoring the model's dynamic and contingent nature. Finally, it provides a measurement framework for operationalizing and assessing organizational capabilities and social

entrepreneurial impact in empirical research.

The SCALERS model builds upon the integration of key theories in management and entrepreneurship. It contends that social ventures, similar to for-profit ventures, must develop and acquire essential resources for growth, adhering to the resource-based view of strategic management (Shelton, 2005; Wernerfelt, 1984). By leveraging their organizational capabilities, social enterprises can create, enhance, and maintain various resource types (Bloom & Smith, 2010). Furthermore, the SCALERS model draws on ecosystem theory, suggesting that scaling social impact involves diverse stakeholders within the entrepreneurial ecosystem, such as governments and the public (Diaz Gonzalez & Dentchev, 2021; Sharir & Lerner, 2006). These dynamic situational factors necessitate adaptive approaches for successful scaling (Bloom & Chatterji, 2009). Effective scaling strategies, therefore, require a strategic alignment – or “fit” – between internal organizational capabilities and external situational factors, embodying the fundamental principle of contingency theory (Linton, 2014).

In this study, the SCALERS model guides the selection of variables, including condition variables, outcome variables, and situational factors. Its emphasis on contingency and the potential for synergistic effects provides a theoretical basis for examining the causal relationships between organizational capabilities and scaling from a novel configurational perspective.

2.2. Scaling up

In social entrepreneurship, “scaling up” refers specifically to scaling the social impact of an organization, rather than its size or revenue in the traditional sense (Desa & Koch, 2014; Shepherd & Patzelt, 2022). It is defined as an ongoing process of “expanding or adapting an organization's output to better match the magnitude of the social need or problem being tackled” (Desa & Koch, 2014, p. 148). Scaling comprises two dimensions: “scaling wide”, which involves extending offerings to a larger audience, and “scaling deep”, aimed at achieving more significant outcomes for each individual (Bloom & Chatterji, 2009, p. 117). Accordingly, the impacts resulting from them are identified as “breadth impact” and “depth impact”, respectively. Breadth impact relates to the scope of an organization's influence, reflecting the extent of its beneficiary base and geographic coverage. Depth impact, conversely, concerns the intensity of the organization's impact, signifying the enhancement of its solutions and the effect on each beneficiary (Desa & Koch, 2014). In essence, breadth impact represents quantitative changes, while depth impact represents qualitative changes (Islam, 2020). Social impact is a central outcome variable in social entrepreneurship research, and it needs to be measured comprehensively by both quantitative and qualitative dimensions (Bacq & Eddleston, 2018; Islam, 2020).

2.3. Organizational capability

Organizational capability is widely recognized as an organization's capacity to effectively deploy diverse resources to execute a spectrum of activities or tasks (Amit & Schoemaker, 1993; Helfat & Peteraf, 2003; Inan & Bititci, 2015). As per the resource-based view, the overarching aim behind developing and leveraging organizational capability is to achieve desired outcomes, often with a central focus on gaining a competitive advantage (Collis, 1994; Ulrich & Lake, 1991). Although social enterprises may not prioritize gaining a competitive advantage over their rivals, they actively cultivate organizational capabilities to better navigate resource constraints and scale up their impact (Bacq & Eddleston, 2018; Day & Jean-Denis, 2016; Desa & Basu, 2013; Morris et al., 2001).

According to the SCALERS model, the organizational capability of social enterprises comprises seven distinct but interrelated capabilities: staffing, communicating, alliance-building, lobbying, earnings generation, replicating, and stimulating market forces (Bloom & Chatterji, 2009). These capabilities are considered essential drivers for the growth

and scaling of social entrepreneurship organizations (Bacq & Eddleston, 2018). To empirically test the SCALERS model, Bloom & Smith (2010) developed formative items to measure these capabilities and validated a positive correlation between each capability and scaling performance, using survey data from over 500 American social enterprises. Subsequent research in diverse contexts, in both developed and developing economies, has consistently reaffirmed this relationship using both qualitative and quantitative methodologies (Cannatelli, 2017; Gauthier et al., 2019; Yu et al., 2022; Zainol et al., 2019). Therefore, the relevance of SCALERS organizational capabilities to scaling up is not only theoretically justified but also empirically validated, fully satisfying the QCA criteria regarding the theoretical or empirical relevance between condition variables and outcome variables.

The definitions and measurement criteria for each SCALERS capability are summarized in Table 1, drawing on the foundational work of Bloom & Chatterji (2009) and Bloom & Smith (2010). Based on these definitions, the capabilities can be grouped according to function or purpose into three broader dimensions. First, earnings generation and staffing are fundamental as they secure the necessary financial and human resources for organizational development, thus belonging to the internal *Development* dimension. Second, communicating, alliance-building, and lobbying involve connections with external stakeholders to acquire resources or form beneficial partnerships, categorizing them under the *Connection* dimension. Finally, replicating and stimulating market forces are methods through which organizations directly expand

their impact by extending their programs or initiatives, fitting into the *Expansion* dimension. Beyond the functional overlap, Bloom & Chatterji (2009) suggest that these capabilities interact synergistically. For instance, strong earnings generation can enhance the effectiveness of all other capabilities, and strong communicating capability can facilitate lobbying and alliance-building. Fig. 1 illustrates both the functional classification and the interplay among SCALERS organizational capabilities, as derived from the literature review.

2.4. Government attention and public attention

Government attention captures the extent to which specific issues occupy the consciousness and agendas of governments (Fan et al., 2023; Simon & Barnard, 1959). It determines the priority of issues in public policy-making (Mortensen, 2010). Studies have shown that governments can facilitate the emergence and development of social entrepreneurship by providing supportive policies or financial subsidies to this sector (Chapman et al., 2007; Choi & Park, 2021). However, when scaling social enterprises, government attention to a specific social issue may play a more direct and significant role than the broader political attention to social entrepreneurship. Supportive public policy is recognized as a crucial situational contingency in the SCALERS model that influences the importance and effectiveness of social enterprises' lobbying capability. When social enterprises address social issues that hold higher priority on the political agenda, they tend to gain government support more readily (Bloom & Chatterji, 2009).

Public attention refers to the level of awareness, interest, or focus that the general public or specific demographic groups devote to particular issues, events, individuals, or phenomena at any given time (Webster, 2011). It is also recognized as a crucial situational contingency in the SCALERS model, directly impacting the relevance of social enterprises' communicating capability. Theoretically, when there is already substantial public concern or support for an issue, social enterprises focusing on that topic may reduce their efforts in communicating their missions and strategies to stakeholders (Bloom & Chatterji, 2009).

2.5. Social entrepreneurship in emerging economies

Emerging economies are those in transition from developing to developed markets, typically characterized by high economic growth

Table 1
Definitions and construct items of SCALERS capabilities.

Organizational capabilities	Definitions	Measurement criteria
Staffing	The capability of an organization to fill its employment and managerial positions with qualified candidates, including both paid employees and volunteers.	How difficult is it for the organization to recruit competent individuals for all positions?
Communicating	The capability of an organization to persuade key stakeholders that its strategy for change merits adoption and/or support.	How effectively does the organization persuade close stakeholders (beneficiaries, volunteers, employees, financiers, public) to support and participate in value creation?
Alliance-building	The capability of an organization to establish partnerships, coalitions, joint ventures, and other connections to achieve desired social changes.	How well does the organization recognize and engage in beneficial collaborations for scaling efforts?
Lobbying	The capability of an organization to advocate for government actions that work in its favor.	How effectively does the organization engage courts, administrative agencies, legislators, or government leaders to champion their cause?
Earnings generation	The capability of an organization to create revenue streams that surpass its expenses.	How easily does the organization cover its expenses and fund its activities?
Replicating	The capability of an organization to reproduce the programs and initiatives it has developed.	How effectively can the organization extend or replicate its services and programs without compromising quality?
Stimulating market forces	The capability of an organization to motivate individuals or institutions to participate in market transactions that facilitate desired social changes.	How effectively does the organization demonstrate that market engagement, including consumer and business activities, can benefit from its initiatives?

Summarized based on Bloom & Chatterji (2009) and Bloom & Smith (2010).

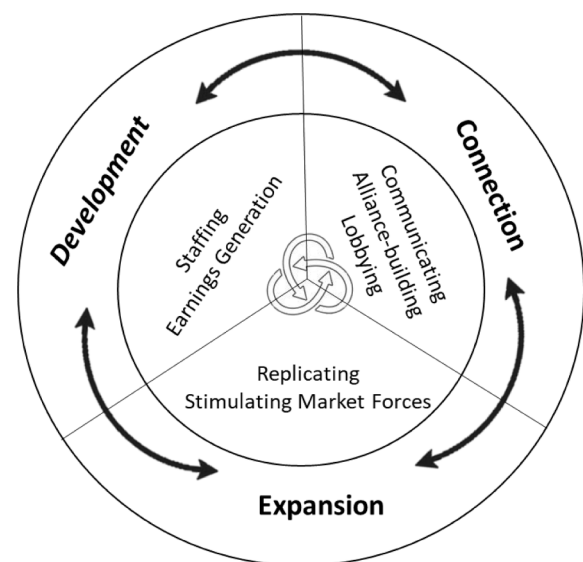


Fig. 1. Organizational capabilities in social enterprises.

rates (Hoskisson et al., 2000). Compared to developed countries, these economies face more complex social challenges stemming from socio-economic transitions, including uneven development and insufficient infrastructure as well as public services (Rosca et al., 2020). However, unlike significantly underdeveloped countries, emerging economies possess dynamic and growing markets that provide the necessary institutional environments for entrepreneurial ventures (Bruton et al., 2008). This unique context makes emerging economies, particularly the relatively more developed regions within them, an ideal setting for social entrepreneurship, which seeks to address societal problems through market-based solutions (Haugh, 2006; Sengupta et al., 2018).

Social enterprises play an important role in bridging institutional voids in emerging economies, notably aiding in poverty alleviation, women empowerment, and equitable education (Rosca et al., 2020). Nonetheless, the characteristic volatility and uncertainty of emerging markets pose distinct challenges for social ventures such as bureaucratic hurdles, lax regulatory frameworks, and unsound financial systems (Diochon & Ghore, 2016; Holt & Meldrum, 2019). In navigating this context, entrepreneurs need to adopt adaptive, non-linear strategies to construct opportunities and achieve entrepreneurial objectives (Rosca et al., 2020).

2.6. Summary: The configurational framework

The SCALERS capabilities are important drivers for social enterprises to scale their impact. Theories suggest that synergies exist among these capabilities and that they evolve as both internal and external conditions change. While existing studies have extensively examined the individual effects of SCALERS capabilities using symmetrical methods, the combined effects of these capabilities remain underexplored. This research gap highlights the need for configurational analyses that go beyond one-to-one causality. Focusing on the unique context of entrepreneurially vibrant regions in emerging economies – characterized by strong social demand and market support for social ventures – this study proposes a configurational framework (as depicted in Fig. 2) to address the main research question: *How can configurations of organizational capabilities effectively drive scaling up in social enterprises?* The research context has considered market-related entrepreneurial factors, but the scaling of social ventures is also shaped by ecosystem actors, such as government

and civil society. Therefore, this study includes a supplementary investigation into how two crucial situational contingencies – government and public attention – might affect the results.

3. Methodology

3.1. Analytical approach

In the field of business and management, Qualitative Comparative Analysis has emerged as a robust tool for examining organizational configurations (Haefner et al., 2023; Misangyi et al., 2017). It assumes that outcomes result from complex interplays among multiple conditions, each contributing heterogeneously to the overall effects (Fiss, 2011). QCA enables the identification of meaningful configurations that lead to specific outcomes, distinguishing between core and peripheral conditions within these configurations (Leppänen et al., 2019; Misangyi et al., 2017). Additionally, QCA accounts for equifinality and asymmetry in causality, making it well-suited for scenarios where different pathways can lead to the same outcome or where causality does not necessarily follow a linear path (Haefner et al., 2023; Kumar et al., 2022; Ragin, 2014). Given the dynamic interplay among organizational capabilities and the research aim to uncover complex causal relationships, QCA is a fitting analytical approach for this study. Since the variables involved are not inherently binary, the fuzzy-set QCA (fsQCA) is specifically employed for fine-grained analysis through a gradational calibration approach (Douglas et al., 2020; Fiss, 2011).

A complete QCA analysis includes case selection, data collection, calibration, analysis of necessary conditions, truth table analysis, solution analysis, interpretation of results, and robustness checks (Leppänen et al., 2019). These steps are elaborated in subsequent sections.

3.2. Case selection

In many emerging countries, including China, social enterprises lack a legal identity and thus do not have unified definitional standards (Sengupta et al., 2018). To ensure the comparability of the cases studied – identified as social enterprises under similar criteria – this research collaborated with the China Social Enterprise Service Center (CSESC), a leading certifier of social enterprises in China. It defines social enterprise

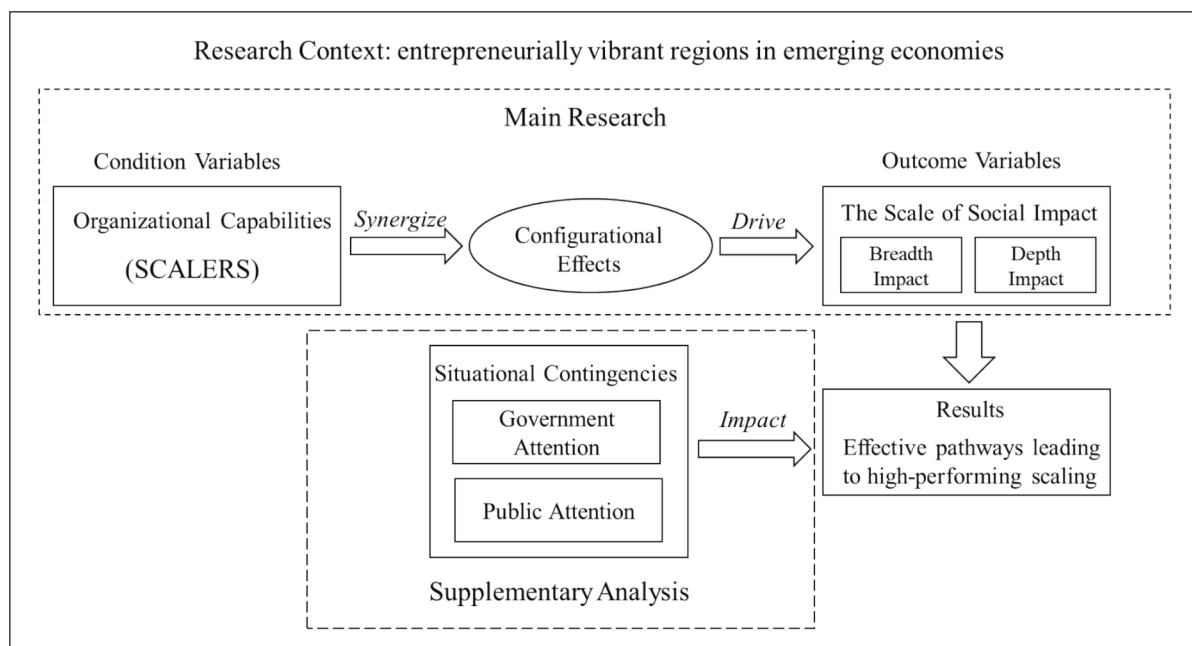


Fig. 2. Configurational Framework.

as an organization committed to solving social problems through innovative business solutions (CSESC, 2022), aligning with this study's criteria. CSESC has certified 598 social enterprises across 27 Chinese provinces, classified into four tiers based on their economic performance and social impact. The research invitation, along with the survey, was sent twice to these social enterprises via CSESC's internal channels in January and February 2024. To encourage participation, social enterprises completing the survey were offered a participation certificate and tailored management suggestions based on the final results. Ultimately, 59 responses were collected, representing a response rate of 9.87 %.

For non-large sample studies, QCA recommends a dynamic, case-oriented selection process that closely considers the unique context of each case (Rihoux & Ragin, 2008). After removing cases without consent for anonymous data usage and those with invalid responses, the pool was narrowed to 52. These cases were then screened through three steps:

- First, their presence in vibrant entrepreneurial environments was assessed using the Index of Regional Innovation and Entrepreneurship in China (IRIEC). The IRIEC, widely accepted among Chinese academics, evaluates regional entrepreneurial vitality across five dimensions: new enterprise creation, inward investment attraction, venture capital attraction, patents awarded, and trademarks registered (Dai et al., 2021; Zhang et al., 2024). According to the latest 2020 IRIEC data, all 52 cases operate in cities scoring over 95, well above the national first quartile of 74.80, confirming their relevance to the vibrant entrepreneurial contexts central to this study.
- Second, cases were evaluated according to the fundamental QCA case section principle: sufficient homogeneity overall and maximum heterogeneity within (Rihoux & Ragin, 2008). This led to the exclusion of two cases due to organization size, one due to lengthy operation time, and another being a Chinese branch of an international social enterprise. This process ensures consistency in key characteristics among the cases. The refined cohort of 48 cases, spanning 13 cities, exhibits homogeneity in size, age, and organizational background while addressing diverse social issues such as disability support, children's education, community development, and elderly care. This diversity indicates significant heterogeneity in the government and public attention received. Additionally, these cases encompass all CSESC certification levels, reflecting

considerable variation in social impact performance and potential organizational capabilities.

- Lastly, two cases were excluded during the review of case data credibility, a process detailed in the subsequent data collection section.

The final set of 46 cases proceeded to fsQCA analysis. Fig. 3 visualizes the case selection process and Table 2 displays the distribution of various organizational characteristics among the sampled social enterprises.

3.3. Data collection, measurement, reliability test

3.3.1. Data collection

This research used a combination of primary and secondary data. Due to the absence of standardized and measurable indicators for organizational capabilities and social impact, these data were collected through a questionnaire survey. The survey was developed based on

Table 2
Organizational characteristics distribution.

Characteristic category	Characteristic classification	Number of cases	Frequency of cases
Organization age	1–5 years	27	58.70 %
	6–10 years	19	41.30 %
Organization size (Number of official employees)	1–20 people	36	78.26 %
	21–50 people	10	21.74 %
Certification level	Tier 1	10	21.74 %
	Tier 2	26	56.52 %
	Tier 3	8	17.39 %
	Tier 4	2	4.35 %
Broad social impact domains	Public service	22	47.83 %
	Support for vulnerable groups	9	19.57 %
	Education	6	13.04 %
	Economic development	6	13.04 %
	Urban development	3	6.52 %

Note: Higher certification levels indicate better performance.

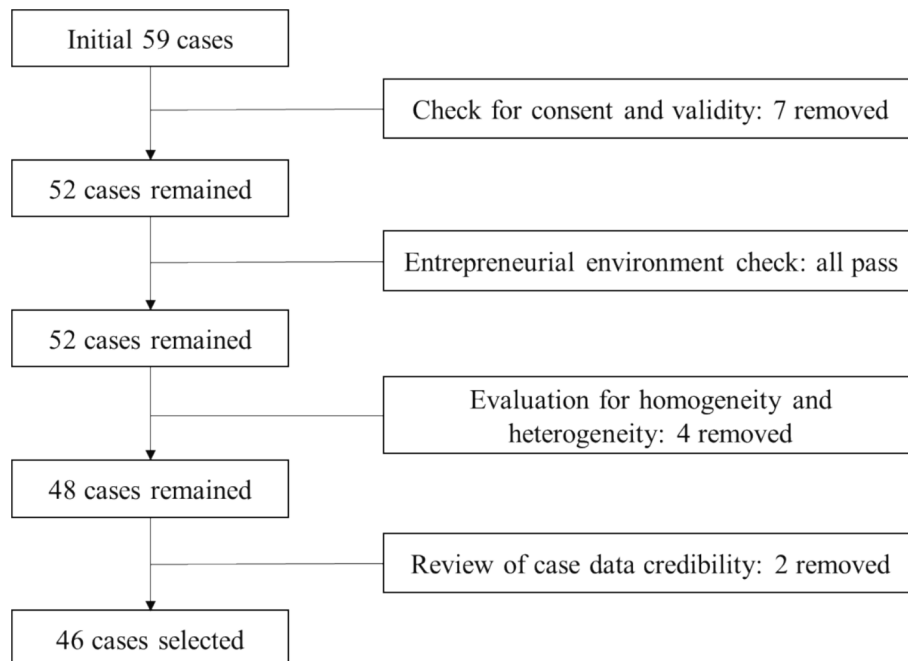


Fig. 3. Case selection process.

Bloom & Smith (2010), who conducted the seminal empirical analysis on the SCALERS model, creating three formative items for each capability and four for social impact. In this study, the questionnaire items were revised to enhance quality and ensure relevance to the Chinese context. For example, the focus of lobbying capability was adapted from advocating for favorable laws to influencing policy-making. The revised questionnaire was reviewed by six researchers specializing in social entrepreneurship and business management. After translation into Chinese, it was further reviewed by three bilingual individuals to ensure linguistic accuracy. Finally, two CSESC experts evaluated the questionnaire to ensure its applicability to the Chinese social entrepreneurship landscape, leveraging their extensive field experience. The complete questionnaire is provided in Appendix A.

Respondents were instructed to rate their agreement with 25 statements on a 6-point Likert scale from 1 (strongly disagree) to 6 (strongly agree). 6 and 7-point scales are believed to optimize information retrieval by engaging respondents' memory and attention capacities (Green & Rao, 1970; Miller, 1956). This research employed the 6-point scale, intentionally omitting a neutral option, to heighten respondent awareness and encourage more deliberate responses through a sensitizing effect (Albaum, 1997). Within this study, higher scores indicate stronger organizational capabilities and more effective scaling of impact. Respondents were advised to consult colleagues for accuracy, ensuring the responses reflect a valid and comprehensive evaluation of the organization (Klein & Kozlowski, 2000).

To ensure the integrity of the data gathered and mitigate potential social desirability bias, several measures were implemented. First, upon sending participation certificates via email, respondents were reminded of the importance of accurate data for this research and prompted to confirm the authenticity of their questionnaire responses. Additionally, responses from each case were cross-referenced with publicly available data from official websites, social media accounts, and news reports to identify any stark contradictions. In cases of potential discrepancies, the researcher directly contacted the concerned companies via email or WeChat to verify the information. These measures led four participating social enterprises to revise their responses, and two cases were excluded due to a lack of response to inquiries about discrepant information.

In addition to self-evaluated survey data, this study incorporated third-party certification results from CSESC to measure social impact. This combined approach enhances the accuracy of the outcome variable, which critically impacts research validity and reliability (Rihoux & Ragin, 2008), and reduces potential respondent subjectivity. Details of this approach are elaborated in the following measurement section. Data for the certification results were sourced from CSESC's social enterprise database.

Data on government attention were sourced from local government annual reports presented by each city's highest official at the annual session of the local people's congress. These reports review the past year's achievements and outline objectives and policy measures for the forthcoming year, covering economic development, social governance, and public services. They are important documents for learning policy priorities and developmental trajectories of local governments (Yang & Zheng, 2022).

Data on public attention were derived from the Baidu Index platform, which is based on user behavior data from Baidu, the largest search engine in China. It analyzes user search queries, treating keywords as units for statistical analysis to evaluate the frequency of each keyword's searches. The Baidu Index is widely used in research to measure the degree of public attention to specific topics (Fang et al., 2021).

To gather data on government and public attention, keywords representing each case's mission and business were identified. Primary keywords were initially extracted by analyzing each case's self-described social goals in the CSESC database. A set of synonymous, searchable terms was then compiled. A research assistant reviewed these terms' relevance by cross-referencing them with the original social goals

descriptions. Any necessary modifications were subsequently reviewed and finalized in discussion with the author.

3.3.2. Measurement

Social impact (SI) is this study's outcome variable, measured through both self-assessment and CSESC evaluation data. In the questionnaire survey, SI is constructed from four formative items, divided evenly between breadth and depth impact dimensions. Formative items are elements that collectively define the concept they measure, with their aggregate value representing the concept (Diamantopoulos & Winklhofer, 2001). Thus, self-assessed SI data were calculated by summing these four items, a method also applied separately for breadth and depth impact. The CSESC certification levels, from tier 4 to 1, were scored as 6, 5, 4, and 3, corresponding to the 6-point Likert scale. Equal weight was assigned to the self-assessment and CSESC evaluation, and the formula for measuring SI is detailed as follows:

$$SI = \left(\sum_{i=1}^n Item_i \right) \times 0.5 + (CertificationScore \times n) \times 0.5$$

In this formula, n represents the number of items that construct a concept, with $n = 4$ for measuring overall social impact, and $n = 2$ when measuring either breadth impact or depth impact. $Item_i$ denotes the quantified value of a respondent's agreement with statement i , which ranges from 1 (strongly disagree) to 6 (strongly agree). Certification Score indicates the score corresponding to each certification level.

The seven organizational capabilities represented by the SCALERS model are the condition variables of this study, with their measurements derived from questionnaire data. The value for each organizational capability was calculated by summing the quantified values of a respondent's agreement on its three formative items.

In the supplementary analysis, government attention and public attention are included as additional condition variables. Government attention toward each case's focus was quantified by averaging the frequency of keywords and their synonyms representing the case's mission in local government annual reports between 2021 and 2023, aligning with the timeframe of the questionnaire survey. Tracking term frequency in government reports is a common method for measuring government attention (Bao & Liu, 2022). Public attention, on the other hand, was measured by the regional average daily Baidu index of these keywords over the same period. For this analysis, both keywords and their synonyms were searched, with the highest occurring value selected for definitive measurement.

3.3.3. Reliability test

The study assessed the questionnaire's internal consistency using Cronbach's alpha coefficient, calculated via Stata/SE software, with results presented in Table 3. A Cronbach's alpha value above 0.7 is typically deemed acceptable (Taber, 2018). The questionnaire shows an

Table 3
Internal reliability assessment.

Variables	Items	Cronbach's alpha
Overall questionnaire	All items	0.918
Social Impact (SI)	(BI + DI)1–2	0.822
Organizational Capability (OC)	(S + C + AB + L + EG + R + SMF)1–3	0.908
Breadth Impact (BI)	BI1–2	0.911
Depth Impact (DI)	DI1–2	0.660
Staffing (S)	S1–3	0.774
Communicating (C)	C1–3	0.892
Alliance-building (AB)	AB1–3	0.907
Lobbying (L)	L1–3	0.883
Earnings Generation (EG)	EG1–3	0.869
Replicating (R)	R1–3	0.772
Stimulating Market Forces (SMF)	SMF1–3	0.837

overall alpha of 0.918, indicating excellent reliability. Specifically, the social impact section scored 0.822, and organizational capability scored 0.908, both reflecting high reliability in measuring these unified concepts. The depth impact variable is the only one with an alpha below 0.7, scoring 0.66. This may be attributed to its formative items – improvement of life quality and optimization of programs – which are less related. Constructs with formative indicators commonly exhibit low internal consistency because the indicators are relatively independent of each other (Diamantopoulos & Winklhofer, 2001).

3.4. Calibration

Calibration is the precondition for QCA, transforming variables into set memberships ranging from full nonmembership (0) to full membership (1) (Ragin, 2009). This research employs the direct method of calibration, using software-based routines to convert numerical raw data into fuzzy-set scores (Mello, 2021). This method requires the pre-definition of three “empirical anchors” for each variable: the fully-out point, the crossover point, and the fully-in point. Based on these anchors, observed cases are classified into one of four areas between 0 and 1: fully out, more out than in, more in than out, or fully in (Mello, 2021; Ragin, 2009). For instance, a social enterprise is considered fully within the high-performing set if its social impact score surpasses the fully-in threshold.

Using percentiles to calibrate continuous variables, particularly quantitative survey data, is a well-established method in QCA, as supported by the methodological literature (Walter et al., 2014). In this study, the quartiles at 25 % (Q1), 50 % (Q2), and 75 % (Q3) were set as the empirical anchors for full nonmembership (0), the crossover point (0.5), and full membership (1) for several reasons. First, quartiles naturally divide the data into four equal parts, ensuring each segment represents a substantial and comparable portion of the dataset. This division enhances robustness, particularly in small to medium-sized datasets. Second, using quartiles helps reduce the influence of extreme values and outliers, offering a more accurate representation of central tendency and dispersion. Lastly, as a common choice for calibration points, the fairness and statistical reliability of quartiles have been demonstrated in many QCA studies (Lou et al., 2022; Sukhov et al., 2023).

According to these cutoffs, values below Q1 were assigned a membership score of 0, values above Q3 were set to 1, and values in between were calibrated through linear interpolation relative to these anchors. Given the confirmed linear relationship between individual organizational capabilities and impact scaling (Bloom & Smith, 2010), linear interpolation is an appropriate method. Table 4 presents descriptive statistics and calibrated thresholds for all variables.

3.5. Data analysis

QCA relies on logic minimization to determine the minimal causal

configurations for a specific outcome, using the truth table as the analysis tool (Duşa, 2018). In this study, the truth table was generated directly using the built-in algorithm of fsQCA4.1 software. The truth table was further refined by setting a case frequency threshold of ≥ 1 and a minimum raw consistency of ≥ 0.8 to ensure configurations are empirically grounded (Du & Kim, 2021; Fiss, 2011; Schneider & Wagemann, 2012). Additionally, the PRI (proportional reduction in inconsistency) threshold was set above 0.75 to confirm the practical validity of the solutions, as recommended in prior studies (Haefner et al., 2023; Misangyi & Acharya, 2014).

To structure the discussion of results and facilitate the identification of potential functional synergies among conditions, this research organized the seven SCALERS conditions into three categories based on the discussion in the literature review: staffing and earnings generation are categorized as *Development* capabilities; communicating, alliance-building, and lobbying as *Connection* capabilities; replicating and stimulating market forces as *Expansion* capabilities.

The fsQCA software provides parsimonious, intermediate, and complex solutions to interpret complex causal relationships, varying the extent of logical reduction (Ragin, 2009). This research chose the intermediate solution to present findings, as it balances the simplicity of parsimonious solutions with the detail of complex solutions, offering an optimal mix of explanatory power and practicality (Glaesser, 2023; Haesebrouck & Thomann, 2021). Additionally, this study used parsimonious solutions to discern core from peripheral conditions within the identified configurations (Fiss, 2011).

4. Results

4.1. Necessity conditions analysis

Before examining causal relationships between antecedent condition configurations and outcomes, it is important to ascertain whether any condition is necessary for the outcome, meaning that it is constantly present whenever the outcome occurs. Identifying necessary conditions allows for their exclusion in subsequent analyses of sufficient configurations, thereby streamlining the model and focusing on the interactions among variables that may produce different outcomes under varying circumstances (Schneider & Wagemann, 2010, 2012). A condition is deemed necessary if its consistency coefficient exceeds 0.9 (Greckhamer et al., 2018). This study found no single organizational capability, whether strong or not strong, is indispensably linked to achieving high or not-high performance in overall social impact or individual breadth or depth impact.

4.2. Sufficient solutions

4.2.1. Configurations for high-performance social impact

In the first round of sufficiency analysis, this research examined overall social impact (SI) as the outcome variable and identified five

Table 4
Descriptive statistics and calibration thresholds.

Variable	Descriptive statistics				Calibration thresholds		
	Min.	Max.	Mean	Med.	Fully out	Cross-over	Fully in
Social impact	14	23	18.18	18	17	18	19.5
Breadth impact	6	12	9.03	9	8.5	9	10
Depth impact	7.5	12	9.15	9	8.5	9	9.88
Staffing	7	18	14.5	15	13	15	16
Communicating	11	18	15.65	15	14.25	15	18
Alliance-building	3	18	14.74	15	14	15	17
Lobbying	3	18	12.39	13	9	13	15
Earnings Generation	6	18	12.5	13	10	13	15
Replicating	6	18	14.15	14.5	13	14.5	16
Stimulating Market Forces	6	18	14.5	15	13	15	16
Government attention	0.67	56	14.56	11.83	6.08	11.83	24.17
Public attention	7	305	105.5	99.5	60	99.5	136.25

configurations of organizational capabilities that drive high-performing scale-up in social enterprises. The results, including the components of each configuration and their consistency and coverage levels, are presented in Table 5, following the guidelines set forth by Ragin (2009). Consistency reflects the strength of causality, and coverage reflects the significance and empirical relevance of the identified pathways. The analysis shows that both the overall solution and individual pathways achieve a consistency coefficient above 0.75, indicating a strong causal link with high-performing scaling (Rihoux & Ragin, 2008).

Adopting the approach of Balzano & Marzi (2023), a clearer graphical representation of the solutions is visualized in Fig. 4, highlighting three distinct patterns among the five configurations. To be specific, configurations SI-1a and SI-1b exhibit identical core conditions, collectively forming Pattern SI-1. Similarly, SI-2a and SI-2b share core conditions, together establishing Pattern SI-2. In contrast, SI-3 is distinguished by a unique set of conditions, thereby singularly constituting Pattern SI-3. Fig. 5 further illustrates these patterns, providing a granular view of the pathways that each configuration encompasses. This visual guide enhances the understanding of the configurations and sets the stage for a deeper analysis of each pathway, framing the subsequent discussion on implications.

Pattern SI-1. This pattern is characterized by the core conditions “ $\sim AB^* \sim L^* R$ ”, which indicate that the social enterprise lacks strong capabilities in forming partnerships (AB) and securing government support (L) but excels in replicating its programs to reach a broader market or serve more beneficiaries (R). In this pattern, the capability to consistently generate revenue that exceeds expenses (EG) acts as a constant peripheral condition. This highlights the importance of financial health and funding in enabling social enterprises to scale their impact through replication, particularly when partnerships are not a strength. With this premise “ $\sim AB^* \sim L^* R^* EG$ ”, social enterprises have two options to scale up effectively by strategically developing other organizational capabilities.

Configuration SI-1a: enhancing communicating and SMF capabilities.

This configuration addresses significant gaps in *Connection* capabilities, ensuring that social enterprises have organizational capabilities deployed across all three dimensions. It also fully activates the *Expansion* dimension, making it the driving force behind the scaling strategy. A typical example of this pathway is Case No. 30, a social enterprise dedicated to rescuing and empowering trafficked women. This organization empowers individuals and generates profits by training rescued women to create and sell handicrafts. Currently, it operates bases in southwestern China and Southeast Asia, demonstrating a strong capability for replication. However, due to the sensitivity of the topic, which faces restrictions in political agendas and public discussions, the organization has encountered challenges in securing government support and forming alliances. Despite these barriers, it has gained steady market support and revenue streams through advocacy and product sales on multiple social platforms. Case No. 30 exemplifies the synergies between communicating, earnings generation, and stimulating market forces, and how they together boost the replication of impacts.

Configuration SI-1b: enhancing staffing capability. This configuration strengthens the internal *Development* capabilities of social enterprises, ensuring a foundation of stable human and financial resources. In this scenario, the social enterprise can replicate or expand its programs into new markets without additional support from external stakeholders in the entrepreneurial ecosystem. A typical example of this pathway is Case No. 8, a platform-based social enterprise with a mission to build a comprehensive soccer ecosystem. Its business includes providing a communication platform for social soccer, building a supply chain management platform for the soccer industry, and promoting urban sports culture. This enterprise operates across more than 70 Chinese cities. Due to the scarcity of similar service providers in China, the company enjoys profitable revenue streams, with stable clients including government agencies, soccer associations, and youth training clubs. The organization's core focus is on developing and operating sports Internet SaaS platforms, making the integration of sports, business, and Internet technology talent a key resource for its growth. With

Table 5
Configurations for high-performance social impact.

	High performance of social impact (SI)				
	SI-1a	SI-1b	SI-2a	SI-2b	SI-3
Development					
Staffing (S)		●	●		⊗
Earnings Generation (EG)	●	●	●	●	⊗
Connection					
Communicating (C)	●	⊗	●	●	●
Alliance-building (AB)	⊗	⊗	●	●	●
Lobbying (L)	⊗	⊗	●	●	●
Expansion					
Replicating (R)	●	●	⊗	⊗	●
Stimulating Market Forces (SMF)	●	⊗		●	●
Consistency	1	0.80	0.94	0.98	1
Raw coverage	0.07	0.03	0.13	0.12	0.05
Unique coverage	0.07	0.03	0.04	0.03	0.03
Overall consistency			0.94		
Overall coverage			0.30		

Note: According to Ragin (2009), ● = core condition present; ⊗ = core condition absent; ● = peripheral condition present; ⊗ = peripheral condition absent; blank spaces indicate “do not care”.

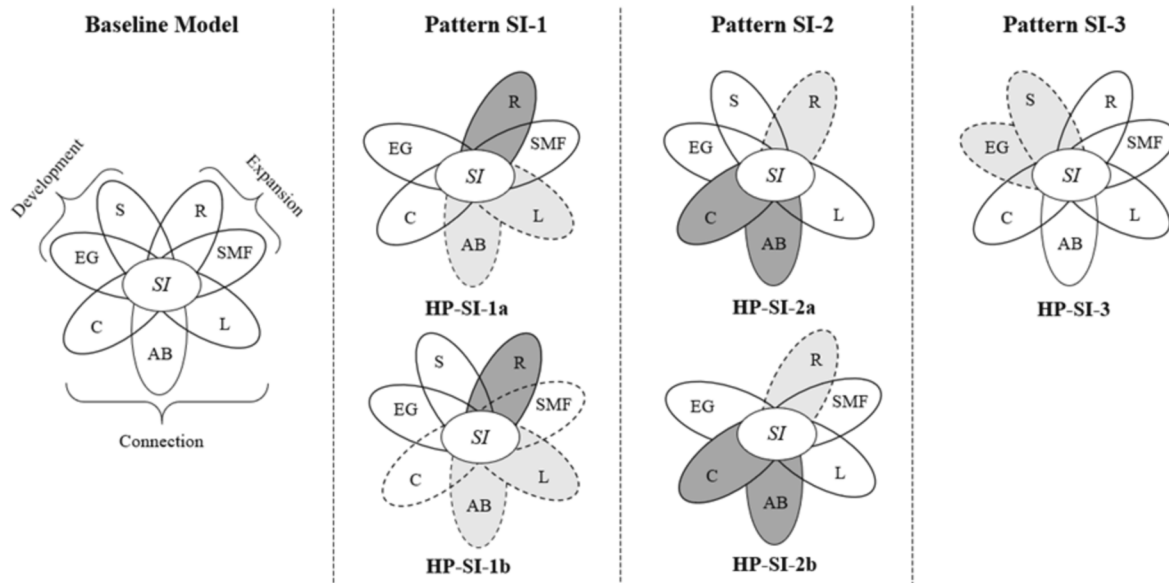


Fig. 4. Graphical representation of solutions for high-performance social impact. Note: Bold lines indicate the presence of a condition, while dashed lines indicate its absence. Dark solid-filled circles indicate the presence of a core condition, and light dashed-filled circles indicate the absence of a core condition. Missing circles indicate “do not care” conditions.

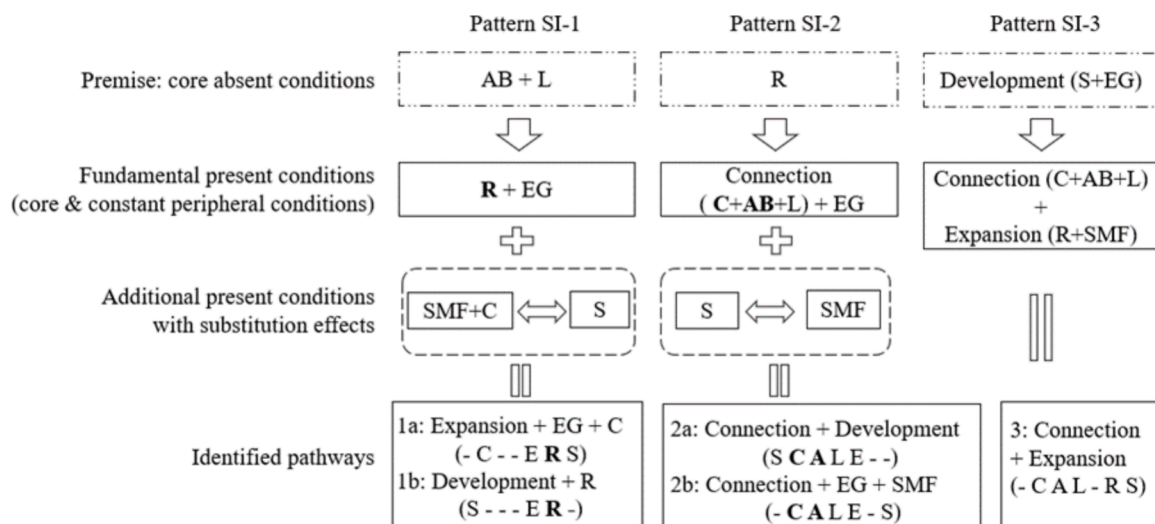


Fig. 5. Configurational patterns and pathways for achieving high-performance social impact. Note: “-” indicates the absence of a capability in the SCALERS model; bolding implies a core condition.

low competition and stable earnings, the company continues to expand its services and impact through talent-driven technological and managerial innovations. Case No. 8 illustrates the synergies between earnings generation and staffing, demonstrating how these elements together contribute to successful replication.

Overall, the SI-1 pattern can be summarized as an “R*EG”-led configuration, which relies on foundational replicable programs and strong profit-generation skills. This pattern is ideal for social enterprises that face challenges in building or enhancing *Connection* capabilities, particularly in lobbying for government support and forming market partnerships. Within this framework, staffing and the combination of communicating and SMF act as substitutes in certain scenarios. Reflecting on the implications of this pattern, this study presents the following proposition within the researched context:

Proposition 1: With limited *Connection* capabilities, social enterprises can scale up effectively by synergizing the enhancement of *Development* and *Expansion* capabilities, focusing on replicating and earnings generation.

Furthermore, the SI-1 pattern reveals that social enterprises with robust capabilities in reproducing their programs are not necessarily dependent on *Connection* capabilities for scaling up. This observation leads to another proposition:

Proposition 2: Social enterprises with easy-to-replicate programs (strong replicating) can scale up effectively by concentrating on enhancing their *Development* or *Expansion* capabilities.

Pattern SI-2. This pattern is defined by the core conditions “C*AB*~R”, and constant peripheral conditions “L*EG”. It indicates that while social enterprises lack strong capabilities to replicate their solutions (R), they excel in *Connection* capabilities (C*AB*L), which include engaging key stakeholders, building partnerships, and securing government support. Additionally, stable financial revenue is essential for ensuring the effectiveness of this pattern. With the setup “~R*Connection*EG”, social enterprises can pursue two strategic pathways to achieve high-performance scaling.

Configuration SI-2a: enhancing staffing capability. This configuration

forges investment in developing *Expansion* capabilities and instead focuses on strengthening the organization's internal *Development* capabilities to leverage strong *Connection* capabilities for scaling. Case No. 10 exemplifies this pathway: a community-based social enterprise dedicated to addressing local issues and improving residents' quality of life. Its services include assistance for lost elderly individuals and preschool education. Due to the low profit margins of these quasi-public goods services, substantial volunteer support and high demand for staffing and revenue generation are essential. Additionally, the enterprise relies heavily on the support of community stakeholders, underscoring the importance of strong *Connection* capabilities. However, because community-based social enterprises often focus on local problems and prioritize improving service quality over market expansion, their programs are less dependent on market support and have low replicability. Case No. 10 demonstrates how the synergy of organizational capabilities within the *Connection* and *Development* dimensions contributes to the organization's success.

Configuration SI-2b: enhancing SMF capability. This configuration leverages market forces, such as business partners or clients, to scale impact by capitalizing on superior *Connection* capabilities and a stable financial foundation. Case No. 5 illustrates this pathway: a social enterprise offering consulting services in low-carbon solutions, sustainable urban planning, and cultural initiatives for governments, corporations, and individual consumers. The company's growth relies on clients' commitment to sustainability, using their platforms to amplify social value. Strong *Connection* capabilities are essential to communicate sustainability's importance, form partnerships that foster sustainable impact, and scale this impact through market-driven channels. Case No. 5 highlights the synergy between *Connection* capabilities and SMF in expanding social impact.

Overall, the SI-2 pattern aligns optimally with social enterprises that have robust *Connection* capabilities, forming a *Connection* (C*AB*L)-led configuration. This pattern particularly benefits social enterprises with context-specific solutions that make replication difficult. Within this pattern, staffing and SMF may serve substitutive roles, enabling social enterprises to choose one or the other based on their unique circumstances and resource needs. Reflecting the characteristics of SI-2 pathways, this study proposes the following within the examined context:

Proposition 3: *With strong Connection capabilities, social enterprises can scale up more effectively by enhancing their Development capabilities.*

Proposition 4: *Social enterprises without easy-to-replicate programs (strong replicating) can benefit from strong Connection capabilities to achieve high-performance scaling.*

Pattern SI-3. This pattern is characterized by the core conditions “~S*~EG”, indicating that social enterprises face internal development challenges due to insufficient staffing at all levels (S) and a lack of stable revenue streams (EG). In this scenario, social enterprises have only one available pathway to achieve high-performance scaling:

Configuration SI-3: developing strong organizational capabilities across both the Connection and Expansion dimensions. Case No. 15 exemplifies this pathway. It is a tech startup social enterprise dedicated to improving life quality for Alzheimer's patients through innovative wearable devices. Due to high costs and risks tied to tech innovation, this company struggles to secure sufficient talent and capital. However, elderly care receives significant public and governmental attention in China, and the market potential is substantial. Consequently, the company is expanding its market reach and impact with considerable support from government start-up subsidies and business partnerships, primarily with nursing homes. This case illustrates how the synergy between *Connection* and *Expansion* capabilities can help overcome shortages in capital and human resources in scaling efforts.

Further comparison of configuration SI-3 to the other four reveals a distinct dynamic: when a social enterprise possesses strong *Development* capabilities, there is a more pronounced substitution relationship between its *Connection* and *Expansion* capabilities. Conversely, in the absence of strong *Development* capabilities, *Connection* and *Expansion*

capabilities exhibit a complementary relationship. Drawing from these findings, the study offers the following propositions:

Proposition 5: *In the absence of strong Development capabilities, social enterprises necessitate strong organizational capabilities in all other dimensions for high-performance scaling.*

Proposition 6: *Strong Development capabilities in social enterprises enable the substitution between Connection and Expansion capabilities for high-performance scaling.*

4.2.2. Configurations for high-performance breadth and depth impact

After examining overall social impact, this research further investigated the specific dimensions of scaling up – breadth impact (BI) and depth impact (DI). Table 6 presents four configurations that lead to high-performance breadth impact and five configurations for depth impact, all with consistency coefficients exceeding 0.75, indicating robust causal relationships. A comparative analysis of the pathways for scaling overall, breadth, and depth impact reveals that the presence and absence of conditions are generally consistent across paths. However, the significance of these presence conditions varies considerably among pathways, as shown in Table 7.

Firstly, the first two configurations leading to high-performance BI and DI, labeled BI-1a/b and DI-1a/b, are identical and mirror the pathways under Pattern SI-1. Yet, within this framework, strong earnings generation (EG) is more crucial for effectively scaling BI and DI than for overall SI, while replicating – central to scaling SI – becomes a peripheral condition for BI and DI. Overall, the configurations under BI-1 and DI-1, along with SI-1 are well-suited for social enterprises lacking strong *Connection* capabilities and instead rely on an “R*EG” combination to achieve high-performance scaling.

The third configuration for BI (BI-2) mirrors the pathway in SI-2b but includes lobbying and SMF as additional core conditions. Similarly, the third configuration for DI (DI-2a) aligns with SI-2a but requires staffing as an extra core condition. These results suggest that Pattern SI-2 is highly relevant for achieving high-performance BI and DI, albeit with added core conditions. Notably, in certain cases, high performance in DI can be achieved with only staffing and alliance-building as core conditions, as shown in configuration DI-2b. This represents a simplification of DI-2a, which is not applicable to scaling SI or BI. This difference may be due to DI's closer connection to the nature of solutions for social problems, enabling social enterprises to scale DI effectively by focusing on talent management or partnerships that enhance and refine their solutions.

The last configurations for BI and DI align with the pathway SI-3 but with specific nuances: BI necessitates all *Connection* conditions as core, while DI mandates all presence conditions to be core. This implies that, in the absence of strong *Development* capabilities, a social enterprise must leverage the full range of other organizational capabilities to effectively scale any type of impact.

Based on the above analysis of configurations that drive high performance in breadth and depth impact, this study proposes the following within the examined context:

Proposition 7: *Organization capability configurations for high-performance social impact scaling are highly applicable to both breadth and depth impacts; however, the core conditions differ by the targeted impact.*

4.2.3. Robustness check

To verify the robustness of QCA results, standard methods include adjusting truth table settings, such as the minimum case numbers and consistency levels, or randomly censoring cases (Schneider & Wagemann, 2012). Two methods were employed in this study. First, the consistency threshold was successively adjusted to 0.75 and 0.85, with subsequent analysis mirroring the initial procedure. At a 0.75 consistency level, six configurations were identified, including the original five, but the overall solution consistency is relatively lower. At a 0.85 consistency level, four configurations were identified, all present in the initial results, with slightly improved overall consistency. Second, 1 to 4

Table 6

Configurations for high-performing breadth impact and depth impact.

	High performance of breadth impact (BI)				High performance of depth impact (DI)				
	BI-1a	BI-1b	BI-2	BI-3	DI-1a	DI-1b	DI-2a	DI-2b	DI-3
Development									
Staffing (S)		●		⊗		●	●	●	⊗
Earnings Generation (EG)	●	●	●	⊗	●	●	●	⊗	⊗
Connection									
Communicating (C)	●	⊗	●	●	●	⊗	●	⊗	●
Alliance-building (AB)	⊗	⊗	●	●	⊗	⊗	●	●	●
Lobbying (L)	⊗	⊗	●	●	⊗	⊗	●	⊗	●
Expansion									
Replicating (R)	●	●	⊗	●	●	●	⊗	⊗	●
Stimulating Market Forces (SMF)	●	⊗	●	●	●	⊗		⊗	●
Consistency	1	0.8	0.93	1	1	1	0.93	0.90	1
Raw coverage	0.08	0.03	0.12	0.05	0.08	0.04	0.14	0.07	0.05
Unique coverage	0.07	0.03	0.11	0.03	0.07	0.03	0.12	0.06	0.03
Overall consistency			0.94				0.95		
Overall coverage			0.27				0.34		

Table 7

Comparison of configurations between different types of impact.

Overall Impact	Effective Configurations of SCALERS					
	—C— ERS	S— ER-	SCALE—	—CALE— S	—CAL— RS	
Breadth Impact	—C— ERS	S— ER-	\	—CALE— S	—CAL— RS	
Depth Impact	—C— ERS	S— ER-	SCALE— A—	S— A—	—CAL— RS	

Note: “—” indicates the absence of a capability in the SCALERS model; bolding implies a core condition.

random cases were censored from the dataset, with analyses following the original procedure. The same configurations were repeatedly obtained, with only a slight 0.02 decrease in overall coverage when 4 cases were removed. The high consistency across different settings confirms the robustness of this study's findings.

4.3. Supplementary analysis of government attention and public attention

In the main research, cases were selected from entrepreneurially vibrant regions in emerging economies, effectively controlling for major entrepreneurial environmental factors such as capital fluidity, technology advancement, and talent availability. To complement this, the supplementary analysis examined how government attention (GA) and public attention (PA) to issues tackled by social enterprises influence the identified pathways. Given the distinct clustering of organizational capabilities, namely *Connection*, *Development*, and *Expansion*, observed in the identified pathways, this analysis consolidated the seven SCALER conditions into these three broader dimensions and introduced GA and PA as the additional conditions. Although this approach neglects the efficacy of individual capabilities, it simplifies the exploration of how new conditions impact existing configurations and mitigates the risks of overcomplexity and overfitting (Rihoux & Ragin, 2008).

Data for the *Connection*, *Development*, and *Expansion* capabilities were calculated by aggregating the scores of the individual organizational capabilities they contain. The calibration approach and truth table

settings were kept consistent with the main study. Prior to analysis, it was verified that neither GA nor PA is indispensable for achieving (not-) high-performance scaling.

As Table 8 displays, four pathways were revealed. The first configuration, SA-1, extends the previously identified SI-2a path by adding high GA, which increases its coverage from 0.13 to 0.28 while maintaining a significant consistency level. This suggests that GA to the issues targeted by social enterprises can positively complement their *Development* and *Connection* capabilities, facilitating scale-up efforts. The second and fourth paths, SA-2 and SA-4, share the same organizational capability conditions but differ in the external attention variables: SA-2 operates without high GA and PA, whereas SA-4 benefits from both. This distinction indicates that the “*Development* * *Expansion*” scaling model is stable and effective across varying levels of external support. The third path, SA-3, shows that in the absence of internal *Development* capabilities, PA more effectively contributes to scaling than GA.

Collectively, these analyses demonstrate that within the scaling pathways identified in this research, GA and PA do not act as substitutes for any of the organizational capabilities but rather serve to enhance them. This complementary role strengthens the effectiveness of the present organizational capabilities in the scaling process. This observation further supports the robustness of the previous findings, indicating that the identified pathways remain stable even with the inclusion of new environmental conditions.

5. Discussion

5.1. Theoretical contribution

This research employs fuzzy-set QCA to analyze complex causal relationships between the organizational capabilities of social enterprises and the scaling of social impact in emerging economies. The findings offer fresh insights into the role of organizational capabilities and scaling mechanisms in social enterprises, contributing meaningfully to key theoretical discussions in social entrepreneurship.

First, this study broadens the understanding of organizational capabilities in social enterprises through the resource-based view (Bacq &

Table 8
Results of the supplementary analysis.

	Configurations	Raw coverage	Unique coverage	consistency
SA-1	Development*Connection*GA*~PA	0.28	0.23	0.86
SA-2	Development*~Connection*Expansion*~GA*~PA	0.10	0.06	0.96
SA-3	~Development*Connection*Expansion*~GA*PA	0.06	0.05	0.89
SA-4	Development*~Connection*Expansion*GA*PA	0.06	0.02	0.93

Note: “~” is the Boolean logical operator for “not”, signifying the absence of the corresponding condition. For example, “~Development” denotes not-strong Development capabilities; “*” denotes “and” in fsQCA software.

Eddleston, 2018; Meyskens et al., 2010). Resource-based view theory traditionally posits that organizations gain a competitive advantage through unique, hard-to-imitate resources, emphasizing the identification and utilization of core resources (Barney, 2001; Madhani, 2010; Mahoney & Pandian, 1992). Much of the research in social entrepreneurship has focused on specific organizational capabilities, such as stakeholder management or networking, which are argued to be central to social enterprises' performance (Cavazos-Arroyo & Puente-Diaz, 2023; Liu et al., 2015; Sacchetti, 2023). However, this study finds that neither any single organizational capability nor any grouped dimension of capabilities is indispensable for achieving high-performance scaling in social enterprises. This suggests that the complexity of social entrepreneurship may require a broader, integrated view of the interdependence of organizational capabilities (Bhardwaj & Srivastava, 2024).

Second, this study contributes to the discussion of contingency theory in social entrepreneurship, which holds that organizational success depends on the alignment between structure, management strategies, and the operating environment (Linton, 2014; Otley, 2016). The use of contingency theory has been increasingly encouraged in social entrepreneurship to analyze how environmental factors shape performance and how social enterprises adapt strategically (Pless, 2012; Short et al., 2009). This research supports contingency theory by demonstrating that success in scaling social impact is contingent on aligning organizational capabilities with situational conditions. The identification of multiple configurational paths to scaling success reinforces the theory's assertion that there is no “one-size-fits-all” approach (Bauwens et al., 2020; Csaszar & Ostler, 2020).

Third, this study explores the synergies between organizational capabilities in social enterprises, a topic that has been infrequently addressed in existing research. The configurations identified in this study reflect subsets of the SCALERS capabilities, within which certain capabilities can substitute for one another. These findings provide empirical support for the hypothesis proposed by Bloom & Chatterji (2009) that not all capabilities need to be fully deployed for successful scaling, underscoring the importance of synergistic interactions. For instance, the study shows that strong *Development* capabilities can enable flexibility in substituting *Connection* and *Expansion* capabilities to achieve high-performance scaling. This suggests that stable human and financial resources can compensate for weaker market forces or limited replicability by strengthening stakeholder connections, or alternatively, enhance market forces and program replicability in the absence of external stakeholder support. These insights on synergies address a gap in the literature, which has largely focused on the separate, linear roles of organizational capabilities in scaling social impact (Bloom & Smith, 2010; Cannatelli, 2017; Yu et al., 2022). They also offer an alternative perspective on the emphasis placed on certain single organizational capabilities, such as profitability or stakeholder management, as essential (Newth, 2016; Sengupta et al., 2018; Urban & Bukula, 2022). However, it is important to note that the results of this study may be influenced by the unique institutional context of emerging economies, where social enterprises often operate within underdeveloped and complex environments (Rosca et al., 2020).

Fourth, this study enhances both the conceptual understanding and practical approach to scaling up by examining its specific dimensions in detail. Comparing configurations that lead to high performance in

different impact types, the study finds a high level of consistency across pathways. This finding questions the prevailing view that scaling wide and scaling deep have distinctly different purposes and thus require differentiated approaches (Desa & Koch, 2014; Islam, 2020, 2022; Moore et al., 2015; Verver et al., 2024). The divergence between empirical results and theoretical perspectives may stem from challenges in clearly distinguishing qualitative and quantitative impact dimensions during the practical scaling process, which can lead to synchronization and overlap in their realization.

Fifth, this study deepens the understanding of social enterprise scaling through the lens of ecosystem theory, which highlights the multilevel interactions between social enterprises and their broader institutional environments, including influences from government policies, market dynamics, and social demand (de Bruin et al., 2023; Roundy, 2017). In a supplementary analysis, this study examines how government and public attention to issues addressed by social enterprises affect the identified pathways. Unlike the suggestions by Bloom & Smith (2010) and Cannatelli (2017), this research finds that public support and supportive policies do not directly replace organizational capabilities such as communicating and lobbying; rather, they enhance and complement these capabilities in emerging economies. This finding acknowledges the important role of government and public support in the social entrepreneurial ecosystem (Bozhikin et al., 2019; Diaz Gonzalez & Dentchev, 2021). However, it also reflects the limited influence of such support on scaling social enterprises in emerging economies, even in entrepreneurially advanced regions – a limitation likely due to the still-developing institutional environment for social entrepreneurship (Sengupta et al., 2018).

Finally, this study extends the application of Qualitative Comparative Analysis (QCA) in social entrepreneurship research. Within entrepreneurship studies, QCA is increasingly recognized for its ability to facilitate subtle inquiries into entrepreneurial dynamics. By aligning with scholars' advocacy for this method (Berger, 2016; Douglas et al., 2020), this study provides a practical example for developing configurational approaches within this domain.

5.2. Practical implications

The research findings provide useful managerial insights for small and medium-sized social enterprises, especially those in developmental stages within vibrant entrepreneurial contexts in emerging markets.

First, prioritizing earning generation capability can be beneficial for social enterprises in most cases. A strong self-development capability, especially in generating revenue, is key in the majority of effective scaling pathways. Financial robustness can grant social enterprises greater flexibility in determining their approach to achieving effective scaling.

Second, once social enterprises achieve stable and sufficient revenue, they may not need to deploy all organizational capabilities. Instead, they can strategically focus on either *Connection* capabilities, to enhance stakeholder interactions, or *Expansion* capabilities, to improve program replicability. This choice can be guided by the ease of developing the required capabilities and the specific demands of the offered programs. For social enterprises with broadly applicable offerings, strengthening internal management to facilitate self-replication and expansion can be

particularly effective. For those addressing highly situational beneficiaries or social issues, it may be more advantageous to refine programs through collaboration with like-minded organizations.

Third, social enterprises, particularly those in early stages of development, may benefit from initially focusing on impact indicators that most contribute to their sustainable growth within their specific contexts, without prioritizing distinctions between breadth and depth of impact. Once their impact becomes stable and their operations more mature, they can strategically reallocate organizational capabilities to either expand programmatic coverage or enhance programmatic quality, aligning with their long-term goals.

Lastly, although a supportive external environment can boost the effectiveness of scaling efforts, social enterprises still need to focus on building their internal organizational capabilities to enhance organizational resilience to better manage the complex and rapidly changing institutional environment.

In addition to the above implications for social enterprises, this research also offers valuable insights for policymakers in emerging economies who aim to harness the power of social entrepreneurship to address societal challenges. Although there are various pathways for social enterprises to scale up their impact effectively, all benefit from a supportive social entrepreneurship ecosystem. Accordingly, policymakers can support the development of SCALERS capabilities within social enterprises through a range of strategic policies. For instance, economic sustainability can be bolstered by offering tax incentives and dedicating a portion of public procurement quotas to social enterprises. Furthermore, governments can take a leading role in establishing platforms that promote collaboration among social enterprises and with other key stakeholders, including large corporations, NGOs, and academic institutions. Developing social entrepreneurship training programs in partnership with educational and professional organizations may help enhance the capacity of social entrepreneurs and their employees. Finally, simplifying the regulatory framework for business registration and licensing can facilitate the establishment and expansion of social enterprises.

6. Conclusion

Scaling is regarded as “one of the most important yet least understood topics” in social entrepreneurship, calling for deeper investigation into its nature and strategies for sustainable growth (Shepherd & Patzelt, 2022, p. 263). This research examines the causally complex relationship between the scaling effectiveness of social enterprises and their organizational capabilities, responding to this research call and offering practical insights for rapidly growing social enterprises in emerging economies. However, the study acknowledges certain limitations that warrant further exploration by future research.

Firstly, this study focuses geographically on entrepreneurially vibrant regions in emerging economies, using samples from China. While this focus allows for an in-depth exploration of social enterprises within a representative context, it limits the generalizability of the findings to other regions and economic environments. The unique socioeconomic and institutional conditions in China may not fully represent those of other emerging economies or developed countries. Practitioners and policymakers should therefore exercise caution when applying these insights in different contexts. Future research should consider broadening the scope by incorporating a more diverse array of entrepreneurial environments and cases from various countries to enhance the findings’ generalizability.

Secondly, the study relies on self-assessments by social enterprises to measure their organizational capabilities, which introduces the potential for respondent bias and subjectivity. While self-assessment provides valuable insights into how social enterprises perceive their capabilities, it may not always accurately reflect their actual performance. Future research could improve the reliability and objectivity of organizational capability measurement by combining self-assessments with objective

data sources, such as third-party evaluations, to provide a more balanced and accurate assessment.

Thirdly, in the supplemental analysis, this study uses *Development*, *Connection*, and *Expansion* – the three broader dimensions of organizational capability derived from the SCALERS model – as new condition variables. Although these groupings are theoretically grounded and show clear clustering in the fsQCA results of the main study, their robustness requires further empirical validation. Future research should aim to validate the empirical robustness of these groupings using larger datasets and different contexts to establish these dimensions as reliable variables.

Finally, the QCA method has limitations in providing generalizable conclusions, given its potential to overlook cases highly correlated with significant outcomes due to sample selection constraints. The primary contribution of this article is to enrich the discussion around the organizational capabilities of social enterprises from a configurational perspective and to present fresh empirical evidence of synergies among these capabilities. To deepen the understanding of these findings and their practical implications, future studies could employ in-depth case analyses to further explore the potential substitution or complementary effects among organizational capabilities observed in this research.

CRedit authorship contribution statement

Yue Xiao: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Questionnaire

Thinking about the last three years* of operations of your organization, please indicate how strongly you agree or disagree with each of the following statements, assuming each statement starts with the following phrase:

Compared to other social entrepreneurship organizations working to resolve similar social problems as our organization....

(* For organizations with less than three years of operation, please provide information starting from your establishment date.).

Scaling Social Impact (SI).

• Depth Impact (DI)

1. We have greatly improved the quality of our solutions to social problems through methods such as program diversification or refinement.
2. We have significantly improved the quality of lives of the individuals we serve.

• Breadth Impact (BI)

1. We have greatly expanded the number of individuals we serve.
2. We have substantially increased the geographic area we serve.

Staffing (S).

1. We have been effective at filling our employee positions with people who have the necessary skills.
2. We have an ample pool of capable volunteers available to help us meet our labor needs.
3. We have individuals in management positions who have the skills to expand our organization, program, or principles.

Communicating (C).

1. We have been effective at communicating our mission (what we want to do) to key stakeholders, including beneficiaries, employees, customers, funders, partners, and the public.
2. We have been effective at communicating our strategies (how we want to do) to key stakeholders, including beneficiaries, employees, customers, funders, partners, and the public.
3. We have been effective at communicating our achievements (what we have done) to our key stakeholders, including beneficiaries, employees, customers, funders, partners, and the public.

Alliance-building (AB).

1. We have been effective at building partnerships with other organizations that have been win-win situations for us and them.
2. We rarely try to “go it alone” when pursuing new initiatives.
3. We have accomplished more through joint action with other organizations than we could have by flying solo.

Lobbying (L).

1. We have been successful at getting government agencies to provide financial support for our efforts.
2. We have been successful at influencing government agencies to create policies, including regulations and rules, that support our efforts.
3. We have been successful at raising the attention of government agencies to the field of our work.

Earnings Generation (EG).

1. We have generated a strong stream of revenues from products or services that we sell for a price.
2. We have cultivated relationships with funders, including donors, investors, and sponsors, who provide us with significant funding.
3. We have been effective at finding ways to finance our activities that keep us sustainable.

Replicating (R).

1. We find our solutions, including programs and models, can work effectively in multiple locations or situations.
2. We find it easy to replicate our solutions, including programs and models.
3. We have been successful at controlling and coordinating our programs in multiple locations.

Stimulating Market Forces (SMF).

1. We have been able to demonstrate that companies can achieve higher revenues or reduce costs by supporting our initiatives.
2. We have been able to demonstrate that consumers can achieve more value or reduce costs by supporting our initiatives.
3. We have been able to trust market forces to help resolve social problems.

Data availability

Data will be made available on request.

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